

"Physics as Information Processing" ~ Chris Fields ~ Lecture 5

Courses/PhysicsAsInformationProcessing_ChrisFields | Chris Fields ~ Lecture 5 | 1:08:41

Hello and welcome everyone. It is September 14th, 2023, and we are in the course Physics as Information Processing. It's lecture five. Looking forward to this lecture and then a little conversation, Chris, off to you. Thank you. Okay. Thank you, Daniel. And welcome to session five of Physics of Information Processing. And today we're going to talk about space-time and particularly about the idea that space-time is emergent from information processing. And this has now become a very mainstream idea in the quantum information and quantum gravity communities. And there's still, of course, some people holding out for the idea that space-time is fundamental. But I would say that at least in these communities, I expect it's become now the dominant idea that space-time is not fundamental, that space-time is not part of the basic ontology of reality or something like that, and that instead space-time is a construct. So that's what today's session is going to be about. So for those of you who are just joining or who would like a little review, we started this course talking about quantum information theory and the idea that quantum information theory is actually a new science about systems that communicate across a boundary. And the systems are always called Alice and Bob for kind of historical reasons. And emphasize that this new science is topological. It's about connectivity between systems. It's not geometric. So it doesn't assume a space-time embedding. So it leaves open the possibility that space-time is actually a construct. It doesn't assume space-time up front. And in the second session of the course, I think, we talked about how this boundary could always be represented as an array of qubits. And that what the systems on each side of the boundary do to communicate is prepare and then measure the states of these qubits. So they communicate by exchanging classical information via a quantum channel. And this introduces a particular kind of quantum noise whenever they're used to the reference frames. So it's not necessarily a perfect communication channel from a classical fidelity point of view. But all of the noise is quantum noise. It's not classical. And it always derives from differences in the reference frames that are used by the two agents.

We reformulated the free energy principle in this setting. And the free energy principle turns out to be very simple. It turns out to be a classical limit of the principle of unitarity, which is just the principle that the way that the information is conserved. And we saw that the approach to perfect alignment between the two agents or perfect prediction on the part of the two agents becomes more challenging, not surprisingly, as system complexity increases. So the more complex the communication, the harder it is to predict, as one would expect.

In the last session, we talked about how constraints on thermodynamic free energy induce compartmentalization in agents. So whenever agents interact with their environments in fairly complex ways,

People like Rudolf Carnap were among the first to actually make precise what this idea of reductive

explanation is.

But the basic idea is that the laws or the dynamics or the formalism that describes behavior at some micro scale, the behavior of little things, actually explains the behavior of big things.

So the basic reductionist idea is that what the little things are doing is what's important.

And what big things are doing are either emergent phenomena or they're just epiphenomenal.

They're essentially just appearances.

And this defines a fundamental scale.

So the real assumption in any reductionist theory is that there's some fundamental scale at which the real dynamics takes place and everything else is just an appearance or it's some sort of emergent phenomenon.

And I try to be very careful when I use the word emergence because it's one of the most ambiguous words in the philosophy of science and people use it to mean all sorts of different things.

But whenever you say emergent, it always sort of implies the idea that it's emergent from something.

And so the idea of emergence is essentially a reductionist idea because it at least implicitly assumes that there's some micro scale from which the macro scale is emergent.

And so the discussions of emergent phenomena are often very consistent with reductionism, even if they claim not to be.

And we're all taught that science is reductionist.

And there are certainly people who publicly claim that science is reductionist.

Richard Dawkins is certainly a shining example of this.

But I suspect that a lot of scientists just kind of play lip service to reductionism without really believing in it because they don't really believe that the macroscopic phenomena that they are studying, life, for example, is completely determined by what's going on at the micro scale or what's going on at the micro scale plus some stochastic noise or something like that.

So I want to explicitly contrast this with a scale-free theory.

And a scale-free theory is not simply a holistic theory.

It's much more precise.

In a scale-free theory, little things and big things obey the same laws or they have the same dynamics or they're described by the same formalism.

So it's not that the macroscale is emergent from the microscale.

The macroscale and the microscale actually obey the same laws.

And the relationship between macro and micro is probably best stated as one of implementation.

Macroscale things have microscale components.

And those microscale components are doing something or other.

And whatever they're doing implements the macroscale system and its behavior.

But this implementation is not an explanation, or at least it's not an explanation all by itself.

And the bumper sticker version of scale-free theories is just as above, so below.

Whatever laws you have at the macroscale are the same ones you have at the microscale and vice versa.

So scale-free theories, in a sense, are simple.

It's the same theory at every level of description.

So what is this idea of scale?

And how does it relate to what we've talked about in this course?

Well, what we've talked about in this course all has to do with what an agent can observe on its boundary.

So scale has to do with how information is encoded on the boundary.

And we can describe that scale in many different ways.

Here's a commonplace way to describe it in physics, by a relationship between energy and distance or size.

And you've probably all heard of the Planck scale.

The Planck scale is the smallest scale at which current physics makes any sense.

And in energetic terms, it's defined by energies on the order of 10^{19} GeV, which is gigaelectron volts.

The LHC, the Large Hadron Collider in Switzerland, gets to about 10^4 GeV.

So many, many orders of magnitude less than the Planck scale.

So the Planck scale is extremely energetic compared to what we can probe experimentally.

The Planck scale energy of 10^{19} GeV corresponds to a distance of about 10^{-35} meters, which is incredibly tiny.

An intermediate scale on this diagram is about the scale of a neutron.

So for example, a proton or a neutron, they have a rest mass of about 1 GeV and a size of about 1 femtometer.

So this intermediate scale is roughly the scale of nuclear physics.

And then if you go to our scale, the scale of things that are about a meter in size, human beings, tables and chairs, lots of animals,

the things we're most familiar with, that corresponds to an incredibly tiny energy.

10^{-5} electron volts is about the energy of a photon of radio, radiation.

And a meter is about the wavelength of long wave radio.

So if you think about visible light, it has a wavelength of about 10^{-6} meters, an energy of about an electron volt.

So it's not surprising that visible light feels energetic to us because it's orders of magnitude above our natural scale of energy,

the sort of the sort of thermal environment in which we exist.

So it's the encoding scale and the boundary that is going to relate theories of things that we see to each other.

And this ends up giving a kind of complicated picture.

If you think of Alice interacting with Bob and measuring things on the boundary...

Sorry, I'm a little bit ahead of myself.

I just wanted to show you a picture of a reductive theory.

This is the Big Bang.

You're all familiar with this.

Something happens at the Planck scale and the entire universe results from it by just a scale change.

So this is a reductive theory.

A scale-free theory is much more complicated.

So here's what a scale-free theory looks like.

Alice interacts with Bob and she's looking at her boundary.

And her boundary states, the boundary states she sees, could be encoded at different scales.

So there may be a small scale, scale one, and a bigger scale, scale two.

And what Alice is looking for is some sort of theory that relates the first scale, the microscale, to the macroscale.

And so it's not that the microscale came first like it did in the Big Bang and the macroscale came later.

These are encodings that can be happening simultaneously.

And the theoretical task is to find theories that relate between these scales.

And these are implementation theories, or you might call them embedding theories from one scale into another.

And what we want as a consistency criterion is for these theories to be constant in time.

So as Bob evolves in time, and so what Alice sees are these encodings on her boundary, she wants the relationship between encodings at different scales to stay constant.

Because if it doesn't, she'll never be able to figure anything out.

She'll just have this chaotic relationship between things happening at different scales, and nothing will make sense.

So you have this complicated picture in scale-free theories.

And if you look at this, and it seems vaguely familiar, or even vaguely, even very familiar, then it should.

Because this is exactly the kind of explanatory structure or semantic structure that you have in computer science.

If you're studying the behavior of a computer, which is just some piece of hardware, some physical system, then you can describe what that piece of hardware is doing at different scales.

And in fact, what we call the hardware level of description, when we talk about our laptops, for example, is typically the level of description of circuits of transistors inside microprocessors.

We think of that as the hardware level.

But of course, that's just a description, too, that's layered on top of lots of much smaller scale dynamics, all the way down to the scale of atoms and eventually nuclei and elementary particles and all that other stuff.

So there's some quantum stuff that's going on that we describe as having a classical hardware level of bit exchange.

And then we describe the behavior of programs on top of that.

So what we want, what makes a computer useful, is that we can assign semantics to programs,

and we can talk about semantic relationships between programs.

For example, the relationship between the internet protocol stuff that Zoom is doing and what you actually see on your user interface and hear on your user interface.

That needs to stay fixed over time,

or Zoom becomes incomprehensible and not programmable,

and in fact, not even well-defined as an algorithmic system.

And the same is true for the relationship between the operating system and the hardware description, or between the operating system and Zoom as a program.

So computer science works in exactly this scale-free way.

And so, in a sense, scale-free explanation is very familiar.

It's exactly the kind of explanation that we have in computer science.

And that's why I refer to the relationship between scales as one of implementation,

because that's a term that we're familiar with from thinking about computers implementing programs.

So what are these theories?

And if we think in biological terms,

we can think of pathways being embedded in a cell,

or cells being embedded in tissues,

all the way up to niches of various kinds,

which may be social niches,

being embedded in ecosystems,

and ecosystems being embedded in the biosphere.

And so we have all of these essentially semantic theories

that talk about how one scale embeds

in another scale.

And if we have these sorts of scale transitions,

we can think of representing the relationship

between transitions at different scales

in terms of the activity of some kind of renormalization group,

which is transforming entities and processes at one scale

into entities and processes at another scale.

And I think in biology,

we can at least put forward the hypothesis

that in all the way from pathways embedding in cells

to ecosystems embedding in the biosphere,

we can put forward the hypothesis at least

that the renormalization group flow

across all of these scales is trivial.

That is that the embedding theories

all have the same formal structure,
that the structure that describes
how pathways embed into cells
is the same structure
that describes how ecosystems embed into the biosphere.

Now, I certainly can't prove that.

I believe that it's likely.

So I think it's an interesting hypothesis to investigate.

And this is an hypothesis that I want to leave you with
as a result of this class is something to think about.

To what extent, in thinking about biological systems
or even social systems,
are the embedding theories the same?

So in thinking this way,
that the physics of the situation
has the same formal structure
as computer science has in terms of explanation,
we're, of course, faced with a problem,
which is that we're not just reconstructing physics.
We're talking about the physics of interacting observers.
We're talking about the physics of communication.

So we're modeling how we do physics.
And that makes our theory self-referential.
And so we have to base the music from Gödel's theorem,
which tells us essentially
that powerful self-referential theories
can't be both consistent and complete.

And by powerful,
I just mean powerful enough to express arithmetic.
So not terribly powerful.

So this becomes an issue that we have to live with.
And we will always be faced with this issue
of having to construct meta-theories,
which tell us how our lower-level theories work.

But we never get to the end of that process.

We never can get to theories
that are both consistent and complete
because our tower of theories

becomes progressively, actually,
more powerful as it goes up.
So Gödel,
in a sense, poses a problem,
but in another sense,
just tells us what life is like as a scientist.
That you can't have everything
all the time
doing science.
But we probably all already knew that.
So let's now talk about space,
end of digression,
and see how these ideas
apply to a system
or a tower of systems
that can construct
a spatial embedding
in its world,
in its observed world.
So let's start out thinking
about a system
that can see its environment,
but what it sees
is a completely uniform environment
where nothing is happening.
So if you have a uniform environment
where nothing is happening,
you can't distinguish
one part of the environment
from another.
So there's no,
you have no sense of space.
So what I want to do
is build up from scratch
what's needed
to have a sense of space.
and then we'll talk
a little bit

about what

organisms are doing.

And I think a very interesting question is

if you look at phylogeny,

at what points in phylogeny,

what lineages in phylogeny

actually need to have

a sense of space?

and what do they have to have

in terms of computational power

to have that sense of space?

So if you just have

a uniform environment,

you don't have a sense of space.

But the first thing

we can do with an environment

is to divide it up

into some number of parts

that are somehow distinguishable.

So suppose we can segment

the environment

into two different parts.

And if you think about *E. coli*,

for example,

E. coli has a bunch of sensors

on the front

and a bunch of effectors

on the back.

So it can sense much more

about the front part

of the environment

than the back part

of the environment.

So already with *E. coli*,

you have this kind

of segmentation

of the environment

into parts.

But so far,
nothing has happened
in this environment.
We just have segments
that can be distinguished.
So let's consider
an environment like this
that has distinguishable segments.
What one next wants
to know about these segments
is whether they're
next to each other.
And if we just have
a bunch of segments,
we just have a set.
And a set doesn't have
any order
or any relationship information.
So I've drawn this environment
as a rectangle of rectangles,
but that's just
a representation.
And it looks like
there are relationships
between these rectangles.
But if they're just a set,
then there are no relationships.
If we want to add relationships,
then we have to add them explicitly.
So let's add some connections
between these different segments
of the environment
to turn that set
into a graph
that says explicitly,
for example,
that the light green segment
is attached

to the light blue segment.

And it's also attached

to the dark green segment.

And the dark blue segment

is attached

to the light blue segment

and attached

to the dark green segment.

So now we have something

that's much more

mathematically complicated

than a set.

We have a graph

that has some

connection information.

So we have a connection

topology.

Now, adding this topology

does something

very important.

It breaks the exchange symmetry

that is there

if you just have a set.

So if we just have a set,

it doesn't make any difference

if I exchange,

for example,

the positions

of the upper right

light blue segment

with the upper left

light green segment.

but if we have

this connection topology,

then it does make

a difference.

Right?

I've

twisted the graph
around.

And if I have
to break connections
to exchange things,
then I've actually
changed the topology.
adding a topology,
adding connections,
can disrupt
a symmetry
that's there
if there is
no topology.

So now let's
think about it.
Now we have the structure
that's required
to think about
something happening.

So suppose
that
at some time
as measured
in the
bit counter
time reference
frame
of the observing
system,
something happens.

So a dot
appears in the
light green segment.

And
we can express
this actually
in the notation

of quantum field theory.

We can think of

the light green segment

as a field

and the dot

appearing

is the occurrence

of an excitation

in that field.

And so we can

write this notation

that says

that the green field

was excited

to produce

a dot,

which is

up arrow

green dot.

And

sometime

after that

it may turn

out that

the dot

disappears

in the green

field.

And

some sort

of dot

appears

down

the dark

blue field.

And

again,

we can

write this
in this
kind of
creation
and destruction
operator
notation
that one
has in
quantum field
theory.

And
so this
says
that
the green
field
is no
longer
excited,
but now
the dark
blue field
is excited.

So this
doesn't
imply anything
about these
excitations
being the
same.

In fact,
they're
completely
different.

One is
an
excitation

of the
green
field
and one
is an
excitation
of the
blue
field.

But at
least now
something is
happening
in the
environment.

So we
can think
about what
does it
take for
an observer
to notice
that something
is happening
in its
environment.

it has
to be
able to
detect
a change
occurring
in some
segment
of its
environment.

And that
change can

involve
something
happening or
something stopping
happening in
that segment
of its
environment.

But it
doesn't
yet have
to have
any idea
of what
an object
is.

But we
can add
that if
the system
has a
memory,
we can
have the
system
connect
these
events
happening in
different parts
of its
environment.

So if
the system
has a
memory that
has some
number of

steps,
say,
n steps,
but here
it could
even just
be two
steps,
it can
see
transitions
between
these two
different
states,
where
there's a
dot in
the green
field,
and then a
dot in
the dark
blue field,
and then a
dot in the
green field,
and then a
dot in the
dark blue
field,
and then a
dot in the
green field,
et cetera.
So I've
expressed this
in this

sort of
field theory
notation
over on
the right
side,
and what
this is
is a
flip-flop.
It's an
observed
periodic
behavior.
So this
is the
basic idea
behind a
clock.
So any
observer that
can see
this,
number of
time steps
internally,
can interpret
what's going
on in
its external
environment as
a clock
ticking.
It can
see some
periodicity.
So as
long as you

can have
memory and
the ability
to detect
change,
you can see
periodic behavior
in the
environment.

so you
can see
the
environment
acting like
a clock.

So let's
think a bit
about what
this does
and does
not require.

So seeing
a clock
in the
environment
requires
having
distinguishable
non-exchangeable
sectors of
the environment.

So you
have to have
this connection
topology
that breaks
the exchange
symmetry.

memory, it
requires having
these sectors
implement some
sort of flip-flop,
so something
periodic
happening, and
you have to
have enough
memory to
see that
periodicity.

But it
doesn't require
any sense of
object identity.
activity.

So the
excitation in
the green
part and
the excitation
in the blue
part can
still just
be field
excitations of
the green
part and
the blue
part.

It doesn't,
there's nothing
that says that
those have to
be the same
thing, they

just have to
appear and
disappear in
some periodic
way.

And having
a clock
doesn't require
a metric
space.

You don't
need to have
any sense
of distance
defined on
the environment.

You just
have to have
this segmentation
that is not
rearrangeable.

So this
tells us
something
extremely
important,
which is
that time
is more
fundamental
than space.

Time in
the exterior
environment is
much simpler
and easier
to observe
than space

in the
exterior
environment.

So if we're
thinking
biologically
about
phylogeny,
we might
expect to
see
organisms
that can
detect
periodic
behavior in
the external
environment
long before
they can
detect
spatial
organization
in the
exterior
environment.

And we
know that
organisms,
even down
to microbial
mats, can
respond to
the diurnal
cycle.

So we
know that
organisms, in

a sense, can
detect
environmental
clocks.
Even organisms
that we have
no reason to
believe divide
the environment
up into
spaces or
the sort of
space that
we're used to,
like a
Euclidean
space.
So now
we're
about adding
space to
this idea
of time
and the
environment.
movement.
So the
next step is
to be able
to see
motion.
And being
able to see
one thing
move from
one segment
of the
environment to

another segment,
as opposed to
something happening
in one segment
and then
something happens
in another
segment.

one has to
have this
idea of
object
persistence,
that there's
a thing
that has
an identity
and its
identity
doesn't
change over
time.

And it
doesn't change
when you move
it someplace
else, when it
appears someplace
else in the
environment.

it's the
same object.

So this
is an
idea that
psychologists
call object
persistence,

and it's an
idea that
physicists call
translational
invariance.

Translational
invariance
means two
things.

It means
that if I
have an
object and I
move it to a
different location,
I don't change
the object.

It's the same
object that I
move from here
to there.

So the
location doesn't
change the
object.

And it also
means the
reverse of
that.

It means that
the object
doesn't change
the location.

My
container
remains
invariant even
though I move

the object
around in
it.

So these
are kind
of the
two faces
of
translational
invariance.

And the
two faces
of object
persistence.

As the
object moves
through time,
its identity
remains fixed,
and it
doesn't change
the time
that it's
moving through.

So if I'm
measuring its
motion with
a clock,
moving the
object doesn't
change the
clock.

These are
clearly very
classical
ideas.
and we
talked about

earlier
classical
communication.

And when
we talked
about
classical
communication
in, for
example, the
context of a
Bell EPR
experiment, we
talked about the
observers being
able to write
down some
data on a
piece of paper
or on a
disk drive or
something, and
hand those
data to
someone else.

So this is
translational
invariance.

The data can't
change when
Alice gives it
to Bob.

Otherwise, there
isn't classical
communication taking
place.

And of course,
there's an up-to-noise

built into here.

The piece of paper
might get wet, and
the ink might run a
little bit.

But it's still
the same
message.

There's still
some identity
being assumed.

So object
persistence is
not an idea we
get from
quantum theory.

It's an idea
that we impose
on top of
quantum theory.

And it's this
idea that's
intrinsically
classical.

So we're
looking right
here at the
quantum to
classical transition.

This is it.
this is what
gets added to
quantum theory
to get
classical physics.

It's this
idea of
object persistence.

Now, once we
have object
persistence, we
can kind of
go to town
constructing
space.

And
here's how
we do it.
we all
learned about
object persistence
this way
when we
were six
months old,
doing motor
babbling,
and
observing our
hands, and
feeling how our
bodies worked,
and seeing
objects that
we could
manipulate,
and that
didn't change
when we
manipulated them.

So, this
is the
birth of
object
persistence.

This is

the birth
of
classicality
in human
psychology.

So, now
let's start
thinking about
building a
coordinate
system.

What's a
coordinate
system?

A coordinate
system is
just a set
of labels
that are
attached to
these segments
of the
environment.

And to
build a
coordinate
system, you
have to
have an
origin.

an
origin
means an
object,
something
that's
fixed,
something

that
doesn't
move
around.
And so,
a coordinate
system
requires the
ability to
identify a
stigmatic
memory,
requires the
ability to
find something
in the
environment
that has
a meaning
that stays
constant over
time.

and that's
what a
stigmatic
memory is.

It's
something in
the environment
that provides
us with a
memory resource
that we can
count on
being the
same.

And it's
having this

fixed point
that allows
us to
label these
connected
segments of
the environment
that contain
objects that
move and
be able to
say,
okay, the
object moved
from this
part of the
environment to
another part,
and I can
give those
parts labels
with respect to
this other
part that
doesn't move,
this origin
point.

And once I
have this,
I can
construct other
invariances.

So for
example, I
can consider
an object
that rotates
around this

origin without
changing its
identity.

And that
defines what's
called rotational
invariance.

And I can
also talk
about an
object that
gets bigger
and smaller
compared to
this fixed
point.

and that
defines radial
distance.

And if you
think about
the baby
in the
previous slide
that's doing
motor babbling,
it sees things
getting smaller
as it moves
them farther
away.

So the
visual system
does this
computation
for us
and constructs
this radial

coordinate for
us with
respect to
our bodies
or specifically
our heads,
our eyes,
body.

So in
doing this
very early
motor
activity,
we're actually
building the
spatial coordinate
system for
our brains
that's implemented
by the
hippocampus,
the place
cells in the
hippocampus.

And if I
have an
origin point
in the
environment,
I can also
infer that
I am
moving,
that I
have changed
my location
with respect
to that

object.

And

that's

a considerably

more complicated

inference because

you have to

have a

representation of

the self,

of you,

to say I

moved.

So this is

something that

comes after

you've developed

this spatial

coordinate system.

you have to

be able to

locate,

in a sense,

metacognitively,

a particular

object that's

you in that

coordinate system.

So we're going

to set that

aside and

now talk

about

adding some

numbers,

so adding a

metric.

And adding

a metric
requires another
particular kind
object,
which is a
ruler,
a particular
fixed object
that can be
moved around
and that has
translational
invariance and
rotational
invariance up
to some
transformation,
up to some
tensor.

And in
Euclidean space,
that tensor is
the identity.

So in
Euclidean
space,
you have
perfect
translational
and rotational
invariance.

As you get
into general
relativity,
then the
ruler can
actually change
its length

as you rotate,
or the ruler
can change
its length
as it moves
through space.
So effectively,
your space-time
can deform,
the coordinates
of your space-time
can deform,
and that gives
you the
representation of
mass,
of course,
in general
relativity.
So we
talked several
sessions ago
about quantum
reference frames,
and I used
the example
of rulers
and clocks
as quantum
reference frames.
So now we
see how to
construct those
from observations
and symmetry
breaking
and object
identity,

all of which
we need
to define
a quantum
reference frame
out in the
environment
like a ruler.

And notice
that we don't
need any of
that environmental
stuff to
define those
QRFs within
our own
processing.

And in fact,
the ones that
exist within
us,
we have no
direct access
to.

We can look
at someone else's
brain and talk
about their
spatial representation
system,
but we can't
look inside
our own
brains and
talk about
our spatial
representation
system.

So this
is how we
get a metric
space
from the
ground up.
and we've
talked about
this from a
very psychological
or biological
point of
view,
starting from
sort of motor
capabilities,
motor babbling,
for example,
a measure of
effort, which
gives you some
sense of mass.
mass.

So if it's
easy to move
something, it's
light, and if
it's hard to
move something,
it's heavy.

And we
understand that
when we're six
months old,
because we can
measure how
much force
we're having to

apply with our
muscles.

We have a
reference
internally for
this force that
we're applying,
and we
use that
internal
measurement to
attribute this
idea of mass
to things in
the environment.

We have to
have memory
for manipulations.

We have to
be able to
identify
variances that
define objects,
object
persistence,
and we have
to be able to
build a ruler
that we can
move around
and use to
define a
metric space.

Now, this is
very different
from the way
physicists
approach this.

If you look
at the physics
literature, then
people start
with information
transfer.

They talk
about entanglement
as a basic
resource.

There's a lot
of discussion
of bulk
boundary
duality, and
the most
well-known
one is the
duality between
any
interspace,
which is the
sort of space
you see inside
a black
hole, and
conformal
field theory,
a quantum field
theory on
boundary of
the black
hole.

So things
you can see,
for example,
going on on
the horizon

of a black
hole are
dual to the
geometry and
the interior
by this
mathematical
relationship.

You can use
that mathematical
relationship to
construct a
quantum error
correcting code
by using the
symmetries of
the field
theory as a
resource for
redundancy that
allows you to
represent the
same thing in
different places.

And as soon as
you can do
that, you can
build up a
network of
elements that
look basically
like elements of
a lattice,
lattice, and
a lattice is a
graph to which
you can assign
some metric

relationships and
you get a
metric space.

And in
physics, typically
the goal is to
get Einstein's
equations out of
this, to start
with some
information
theoretic sorts
of constructs and
get the equations
of general
relativity.

But what are
the equations of
general relativity?

I mean, these
are things that
Einstein derived
by doing fairly
simple thought
experiments using
the invariances
of ordinary
Euclidean space
combined with
ideas from
special relativity
about the speed
of light being
constant.

And he added
the principle of
equivalence, which
is the idea

that the
essentially mass
and acceleration
are the same
thing, or
that gravity
and acceleration
are the same,
that weight and
acceleration are
the same thing.
so it's at least
reasonable to
expect that
this inferential
process that's
going on in
theoretical physics
is generating
the same thing,
although in a
very formalized
way, that we
can generate by
thinking about how
organisms deal with
their environments.

And so I want
to leave this
with you as the
second hypothesis
here.

The first one
was the hypothesis
about a trivial
renormalization group
flow across
scales in

biology.

The second hypothesis is that these two approaches to emergent space-time are actually giving us the same information, and maybe starting out with the same information, in fact, that we may be able to think about these fundamental building blocks dynamics of information flow and entanglement in terms of the sorts of perceptual relationships that we've been talking about in this class.

So there's our second hypothesis to think about going forward. And I just want to leave you with the idea that organisms, at least, manage

this very
interesting
process
that we
can think of
starting with
this construct
of memory.

Organisms have
memory of
multiple scales,
and it's memory
that allows them
to formulate the
idea that there
are objects.

objects.

And as soon
as you have
objects that
persist, you
have a source of
redundancy.

So you can do
error correction.

And if you can
do error
correction, if you
can move the
same thing from
one place to
another and have
it encode the
same information,
then you can
build space
time.

And as soon

as you have
space time, you
can use the
space time as
more memory.

You can use
the space time
itself as an
error correcting
code.

Because space
time has the
nice ability to
hold lots of
copies of some
informative
structures.

So you can build
libraries, or you
can build computers,
or you can build
the internet if you
have space time,
which gives you
more memory, which
gives you more
object persistence,
which gives you
more error correction.

and the circle
just spins
around.

So organisms do
this.

And the question
is, how does it
get off the
ground?

And next
time we can at
least speculate
about that kind
of origin of life
type of question
in the context
of the FEP
and its
realization in
simple systems
of communicating
observers.

So as
always, I
encourage you
to use the
interactive Q&A.

The fifth
discussion session
with Andrew,
who's unfortunately
not here today,
will be the last
day of September,
Saturday at
5, as usual,
5 o'clock
European time,
8 o'clock
California time.

And then the
last session of
the class will
be the 12th
of October.

So thank
you very much,

and I look
forward to your
questions in the
interactive Q&A.

Thank you,
Chris.

Can I ask
some short
questions?

Sure.

All right.

So first,
from Lana,
wrote,
is the
periodicity
the time
lengths
between and
during the
excitations
built in?

So this was
when we had
the four
cells,
and there
was the
inklings of
a clock
coming into
play?

Okay.

I think
the simplest
answer is
that
if nothing

else is
happening
in the
environment
except this
periodic
behavior,
then
it's not
actually
meaningful to
talk yet
about the
time between
events.

So if I
have a
periodic,
a system
that's
quasi-periodic,
for example,
and nothing
else is going
on in my
environment,
then I can't
distinguish that
from a perfectly
periodic system
because I would
have to have a
perfectly periodic
reference to
know that the
other system
was aperiodic.
And if you

think about
the internal
clock being
driven by
information flow
into the
system,
then something
has to be
happening in
the environment
for information
to be flowing
into the
system and
being counted.

So again,
as we talked
about a few
sessions ago,
the internal
memory,
only gets
updated when
something
happens.

So internal
and external
get tied
together.

And this
is why we
can use
external
clocks.

But it's
also why we
depend on

external
clocks,
because our
intuitive sense
of time is
very malleable.
clock.

And so we
actually, as
humans, use
external clocks
to correct for
the malleability
of our own
intuitive sense
of time.

power.

Wow.

And there's
probably a lot
to be said
for that
procrustean
nature of
having the
chronometer,
the clock on
the wall,
and the
tick,
tick,
tick,
as an
evolutionarily
novel ruler
of time
playing a
logistical role

in time-bound
society.

Yeah.

And, you
know, think
of earlier
clocks that
earlier cultures
have used and
that animals
use.

I mean, we
have the
solar clock,
the lunar
clock, etc.

And, you
know, the
orbit of
Venus clock
that was so
important to
the Mayans
and so on.

So there
are lots of
available clocks
in the
environment.

If you have
the ability
to notice
them and
enough memory
to realize
that they're
clocks.

Awesome.

All right.

Here's a

question from

Upcycle Club.

They wrote,

how does

representing the

network structure

and dynamics

with quantum

embeddings

affect the

semantic

interpretation

of the

network?

The

embedding

theory is

the simplest

semantic

interpretation

of the

network.

So,

and again,

a good way

to think of

this is

through the

lens of

computer science.

You know,

as soon as

you have a

programming language

that allows

variable binding,

then you've
built a
very simple
kind of
semantics
into the
language.
If you have
x as a
variable and
you have
variable binding,
then at
different times
in the
execution of
the program,
x can
have different
causal
consequences
for downstream
computation,
because it's
been bound
to a
different
value,
so it
has
different
semantics.
So the
semantics
of x,
again,
go back to
Gregory Bateson.

The basis
of semantics
is differences
that make a
difference,
make a difference
to behavior,
make a difference
to action
ability.

So
variable binding
gives you
differences that
actually make a
difference to
what the
program does.

So
what
is that?

That
is the
semantics
that's assigned
by your
embedding
theory
between what
the programming
language is doing
and what the
operating system
is doing
and so
forth.

And I
think it's

no surprise
that
the
semantics
of programming
languages
end up
being
represented
with these
very general
mathematical
constructs
like category
theory,
which
are
essentially
tools
for representing
arbitrary
relations
relations
between
systems.

And that's
what embedding
theories are.

They're
theories that
state
relationships
between
systems.

awesome.

And very
provocative
to explore

on the
biological
and life
and on
the
space,
time,
and matter
sides,
which are,
of course,
seamlessly
integrated
through,
if only,
our own
experiences.

Yeah,
this is
another
place where
it's
useful to
go back
to this
difference
between
reductive
theories
and scale-free
theories.

And scale-free
theories are
all about
semantics
because they
represent the
existence of

stuff happening
at different
scales.

And so they
naturally raise
the question
of what
are these
relationships?

What are
these embedding
theories?
theories?

And in
reductive
theories,
there's a
very deep
sense in
which semantics
doesn't
matter.

If you
think that
all that's
going on
that's
important is
the stuff
that's going
on at the
Planck scale,
then you
don't need
any semantics.

And so
you see
this in

late
60s,
early
70s,
old-fashioned
AI,
where
people talk
about
computation
as purely
syntactic.
And this is a
very reductionist
view of
computation.
It leaves
out the
semantics.
And without
the semantics,
the computation
is useless.
It doesn't
tell us
anything at
all unless
we can
map it
to something.
So
we have
this
philosophy
of science
tradition that
tells us that
semantics is

not important.

And we end

up with

tools like

the FEP

that essentially

tell us that

semantics is

all important.

from a

biological

point of

view,

the most

natural

response is,

of course,

we knew

that all

the time.

That's why

we're doing

biology.

So all

of this

ties together.

would you

say that

FEP says

semantics

is important

and then

what does

it say

about what

semantics

are important

or is that

entirely
left up
to the
modeler?

Well,
it tells
you a lot
about what's
important
for the
system
itself.

The system
itself is
dealing with
its environment
and it's
trying to
increase its
predictive
power.

So it's
building a
theory of
the environment
and that
theory is
intrinsically
semantic.

It's about
the environment
and it's
actionable.

The theory
is meant
to drive
action on
the environment

that will
either yield
more information
or get
the environment
to behave,
to do
what I'm
predicting
it's going
to do.

So I
think the
FEP is
intrinsically
semantic in
that sense,
that it's
about an
agent modeling
its environment.

Yes,
and about
the semantics
of action
as opposed
to a
partial
or limited
semantics
of passive
inferers
or semantics
of abstraction
which kind
of leaves
the important
link between

the agent
and the
environment
through sense
making and
decision
selection
basically
unaddressed
because it's
kind of
Descartes
in the
thought experiment
in the
room and
here when we
actually engage
with the
boundary
of the
room,
that's
where we
start.

Yeah.

Yeah,
the FEP
completely
destroys
this
classical
notion
of the
passive
observer
as does
quantum

theory.

There are
no passive
observers
in quantum
theory.

So this
is a
construct
that we
inherited
from a
classical
worldview
that is
seriously
misleading.

A lot
more to
say, but
that's a
great place
to end.

Thank you,
Chris.

Okay,
thank you,
Daniel, and
thanks to
you all,
and ask
questions.

See you next
time.

Cheers.

We'll see you
next time.
to

Thank you.